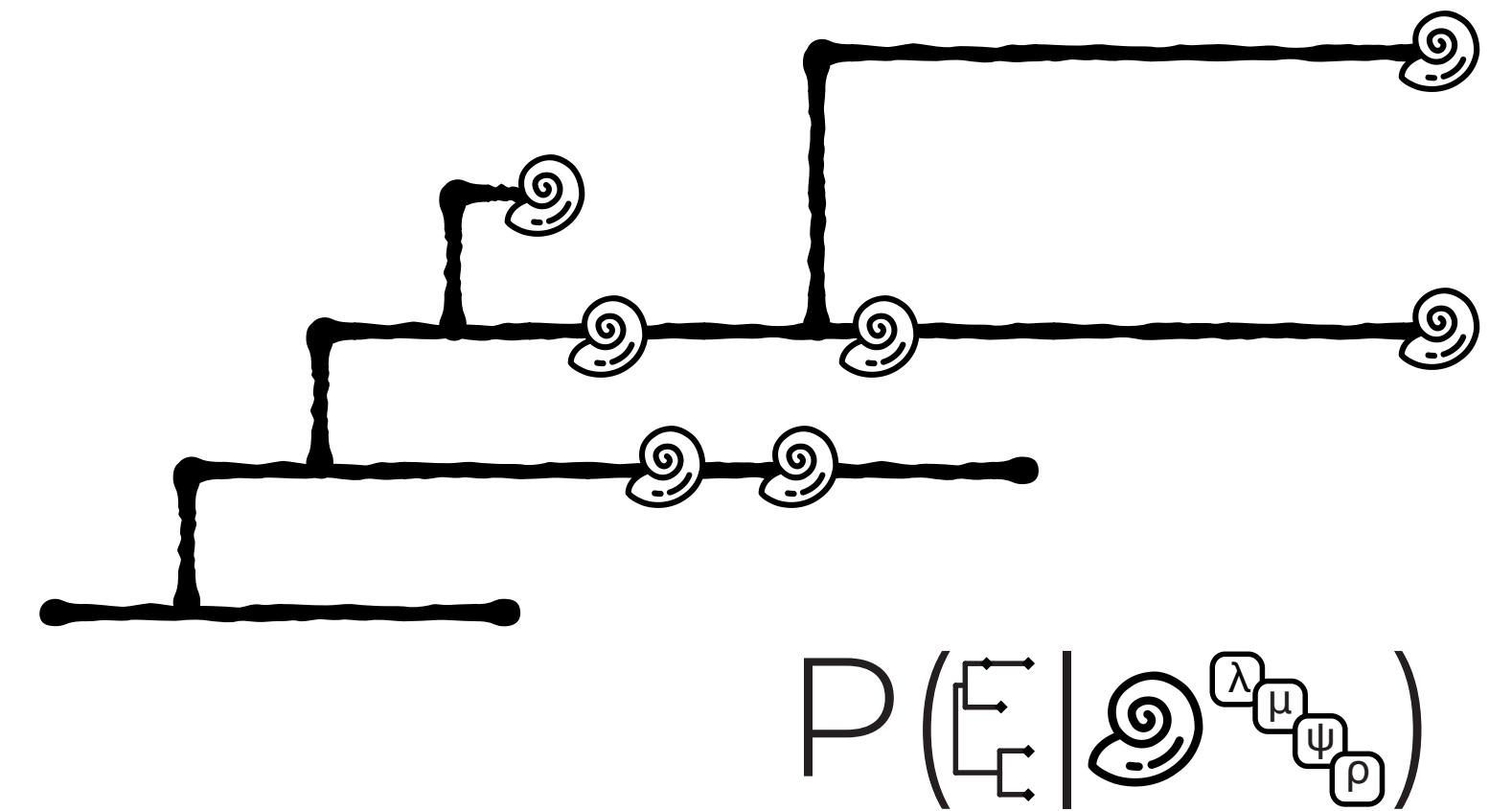
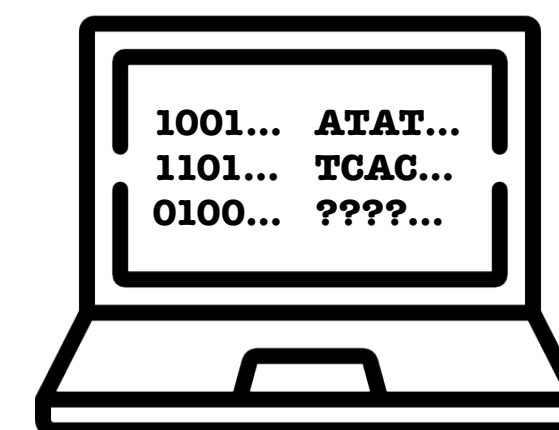
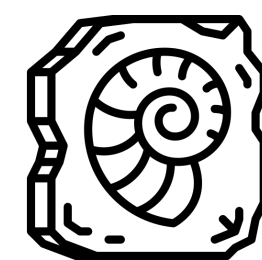


BDMM-Prime for Macroevolution

Taming the BEAST Beijing

Rachel Warnock

17.07.2025





Tim Vaughan

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Author: Tim Vaughan

Created: 2025-07-14 Mon 17:01

BDMM-Prime

Linear birth-death-sampling phylodynamics

Source code: <https://github.com/tgvaughan/BDMM-Prime>

1. Introduction

BDMM-Prime is a BEAST 2 package for performing phylodynamic inference under a variety of linear birth-death-sampling models with/without types.

1.1. Comparison with BDMM

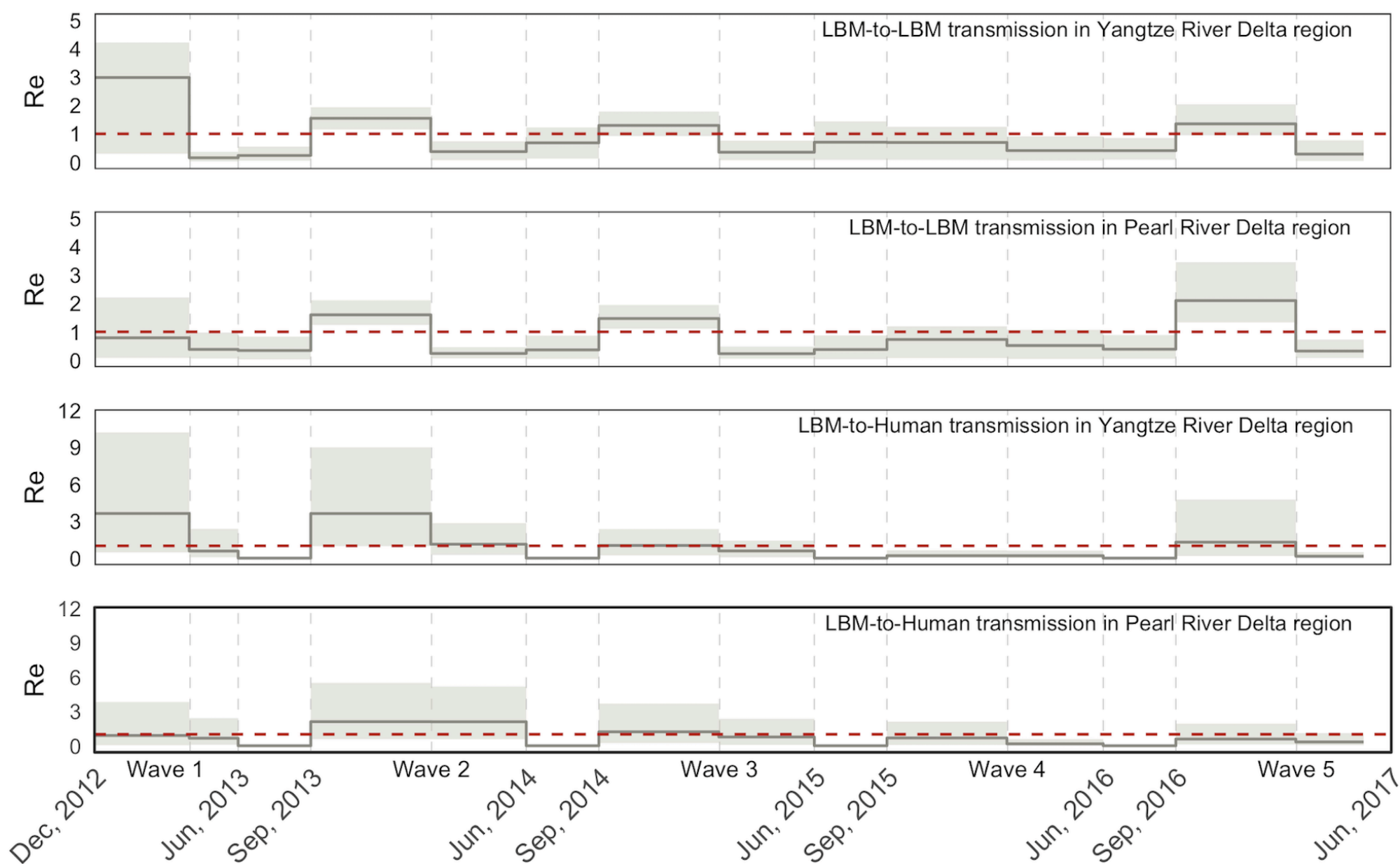
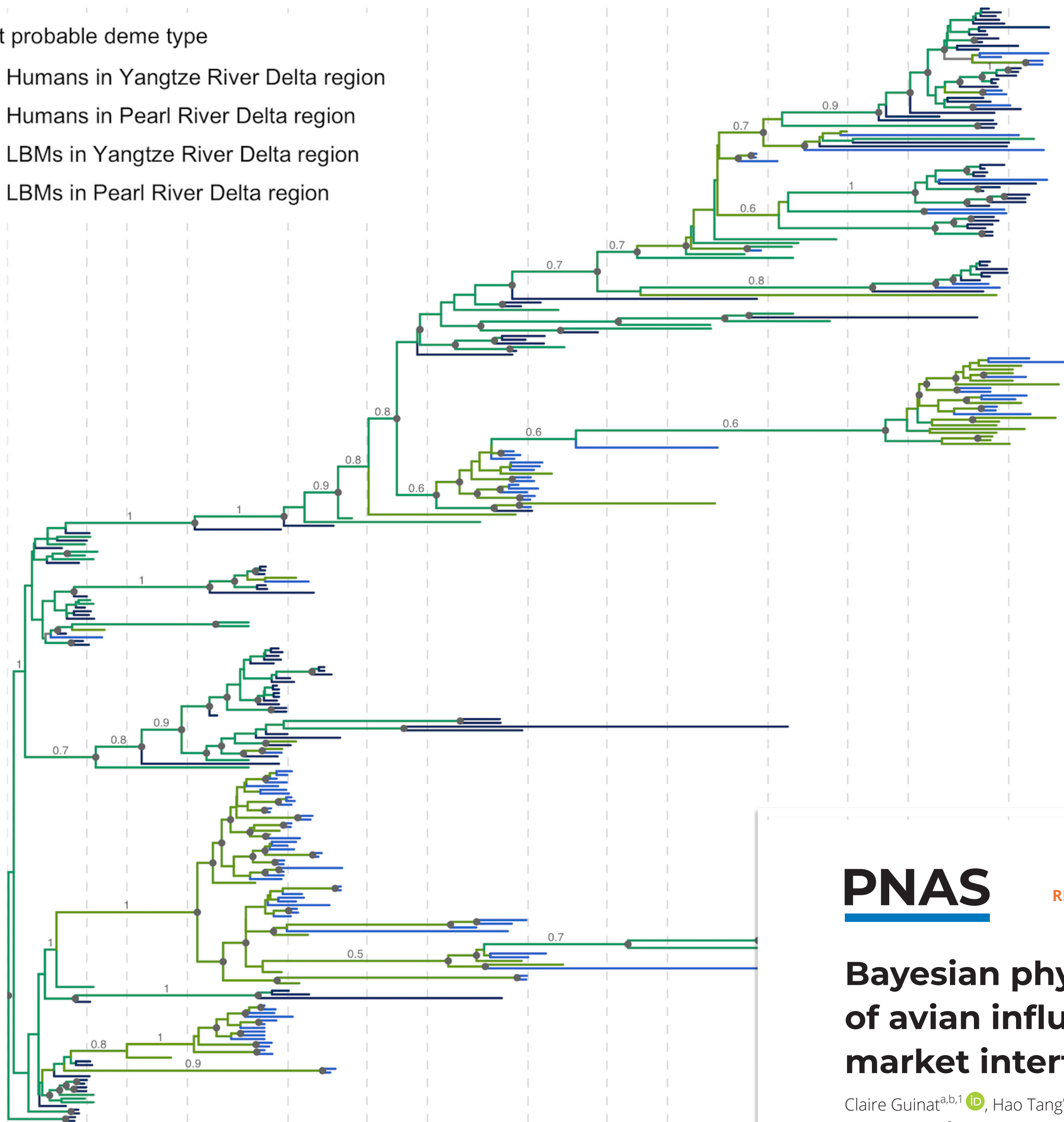
The BDMM-Prime project is a hard fork of the original [BDMM project](#). The intention is to extend the functionality of the original package, while improving its flexibility and ease of use. It incorporates the following enhancements:

- an improved BEAUti interface that allows a much more diverse range of analyses to be configured,
- automatic fall-back to analytical solutions for unstructured (single type) analyses, meaning BDMM-Prime can in principle be used in place of [BDSKY](#),
- use of stochastic mapping for efficient sampling of ancestral types

<https://tgvaughan.github.io>

Most probable deme type

- Humans in Yangtze River Delta region
- Humans in Pearl River Delta region
- LBMs in Yangtze River Delta region
- LBMs in Pearl River Delta region



PNAS

RESEARCH ARTICLE

EVOLUTION

OPEN ACCESS

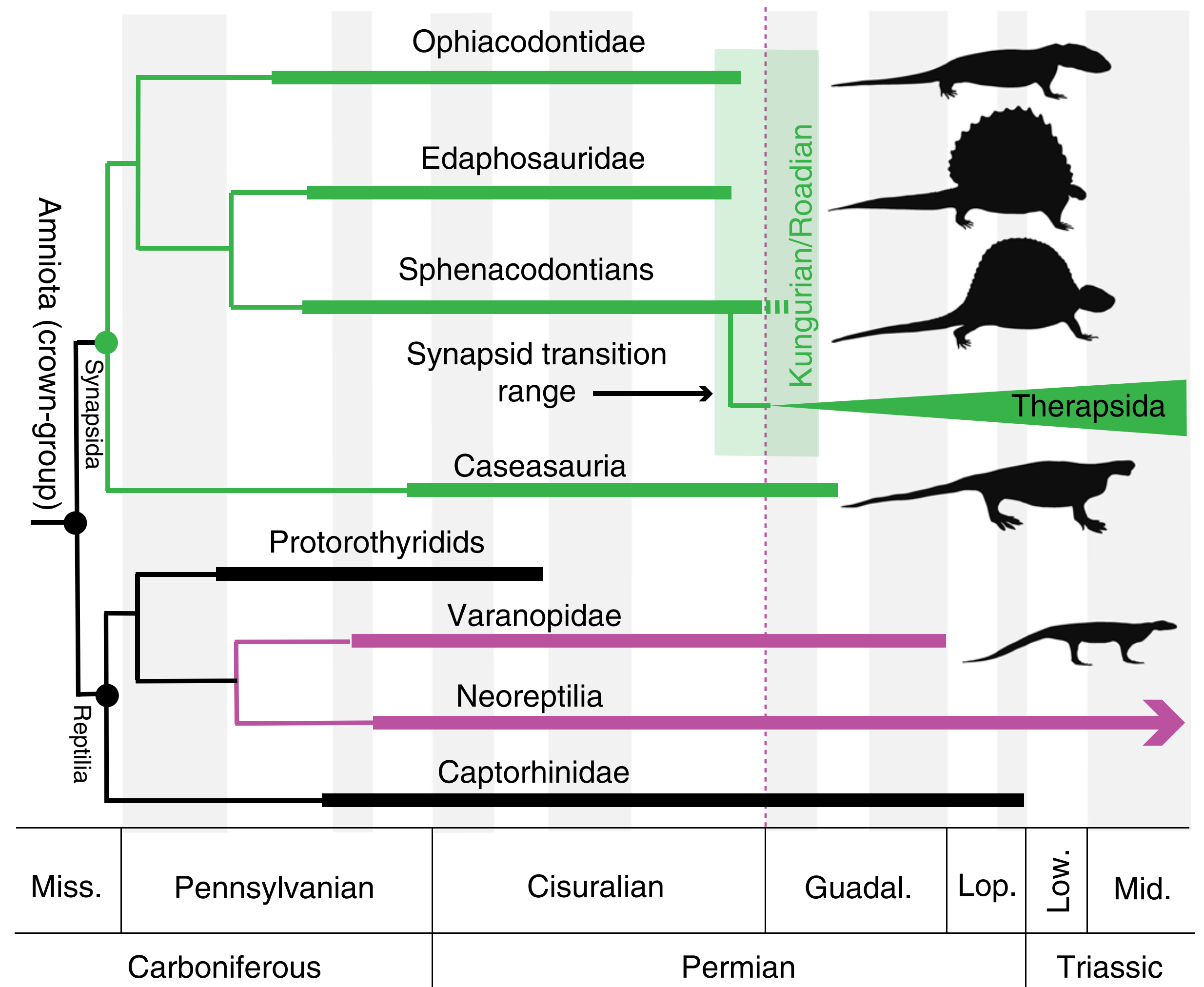


Bayesian phylodynamics reveals the transmission dynamics of avian influenza A(H7N9) virus at the human–live bird market interface in China

Claire Guinat^{a,b,1} , Hao Tang^c , Qiqi Yang^d , Cecilia Valenzuela Agüi^{a,b} , Timothy G. Vaughan^{a,b} , Jérémie Scire^{a,b} , Hongjie Yu^e, Wei Wang^e, Zhiyuan Chen^e, Mariette F. Ducatez^f , and Tanja Stadler^{a,b}

Exercise

Both skyline and multi-type extensions of the FBD model are available via BDMM-Prime

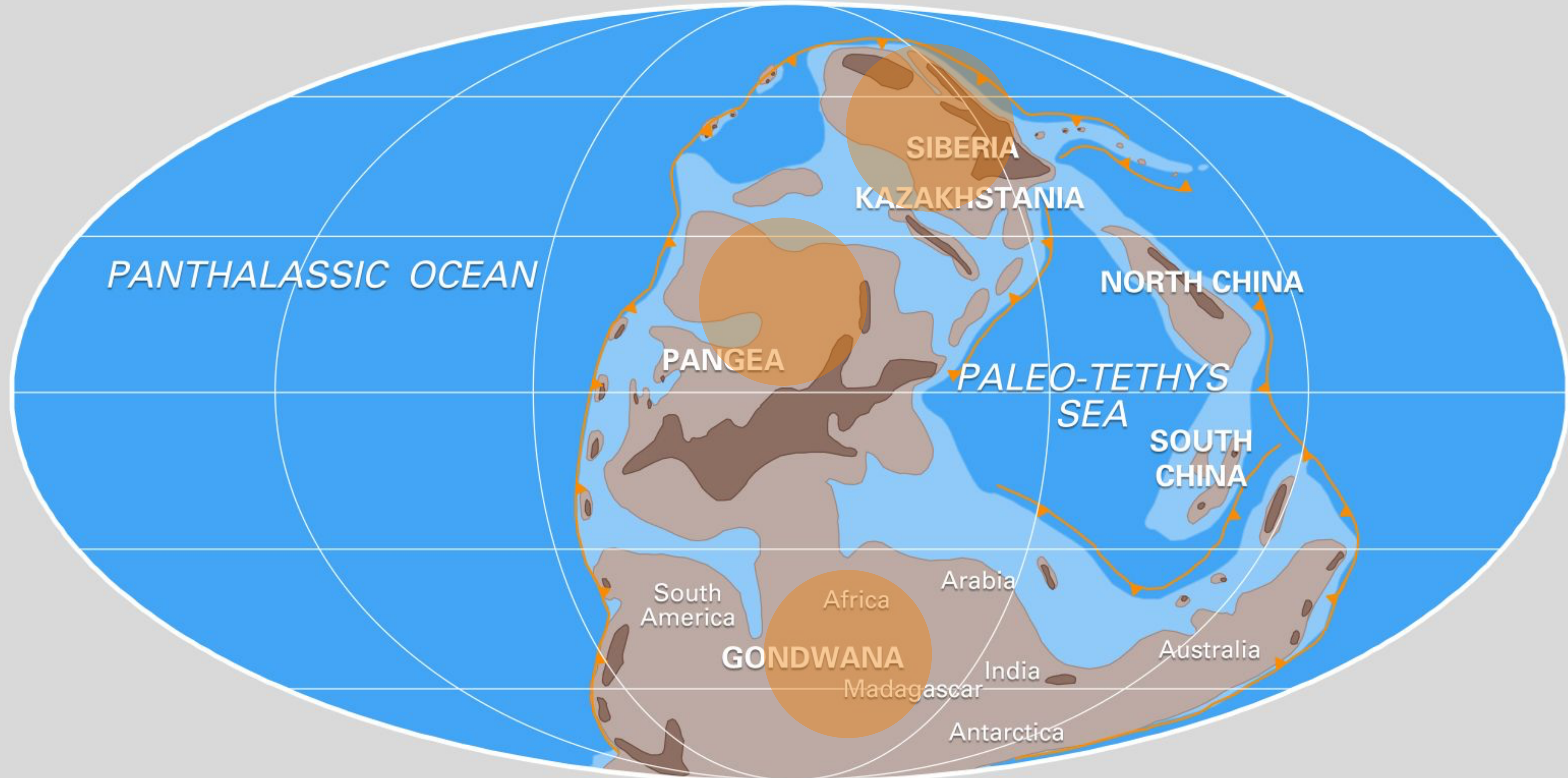


- 70 fossils, spanning the Carboniferous to the Triassic
- Not all lineages are extinct
- 294 morphological characters
- Fossil ages plus locality information

```

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2  [Data written by write.nexus.data.R, Tue Jun 17 20:12:20 2025]
3  BEGIN DATA;
4  DIMENSIONS NTAX=70 NCHAR=294;
5  FORMAT DATATYPE=STANDARD MISSING=? GAP=- INTERLEAVE=NO symbols="0123456789";
6  MATRIX
7  Gephyrostegus_bohemicus_Eu_309      00030000000000-000000?00000-030----0??
8  Seymouria_spp_NAm_288.51           00000000?00000-000000001200-01110000??
9  Tseajaia_campi_NAm_296.9           001-10000001?10001?10?00000-01111000??
10 Limnoscelis_paludis_NAm_298.61      001-1001?00110-001011200000-01100000??
11 Oedaleops_campi_NAm_296.21         100?21-0000111000??11?00100-011111000-
12 Eothyris_parkeyi_NAm_285.05        10??21-00001110001011?00100-01111101??
13 Vaughnictis_smithae_NAm_294.5      ?01-?0000001???????11???????0?????????
14 Eocasea_martini_NAm_300.45         ??012???0???????????????????0?????????
15 Euromycter_rutena_Eu_288.51        100121-0001110-0010002002010010----00-
16 Casea_broilii_NAm_283              100121-0001110-000000?00100-010----00-
17 Ennatosaurus_tecton_Eu_269.025     100221-0001110-0000112002010000----00-
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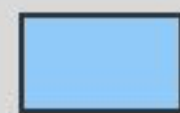

Late Carboniferous 306 million years ago



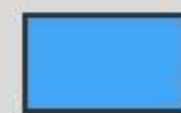
Mountains



Land



Shallow seas



Deep ocean basins

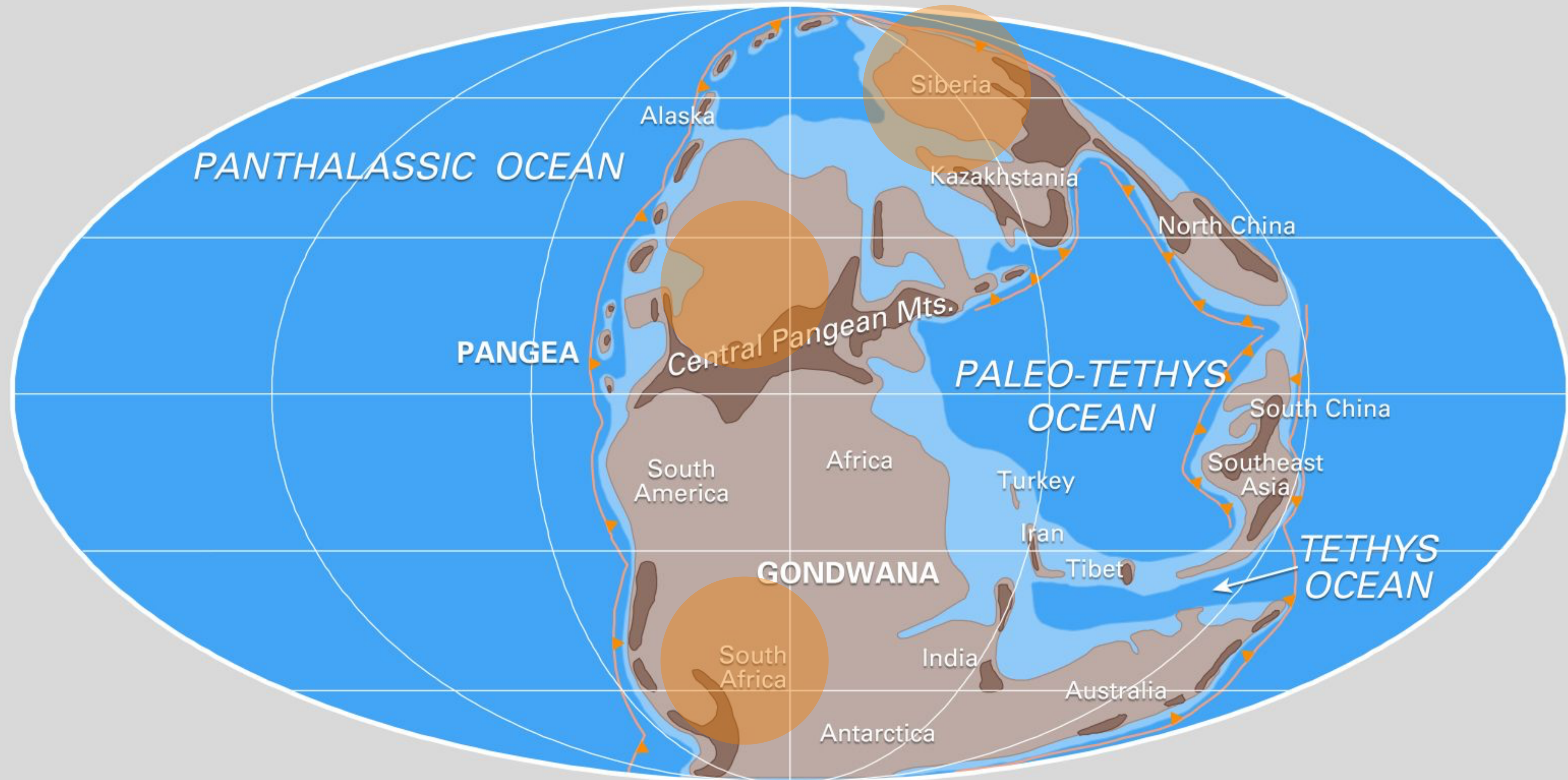


Subduction zone
(triangles point in the
direction of subduction)

SOURCE: © 2001 C.R. Scotese, PALEOMAP Project

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Late Permian 255 million years ago

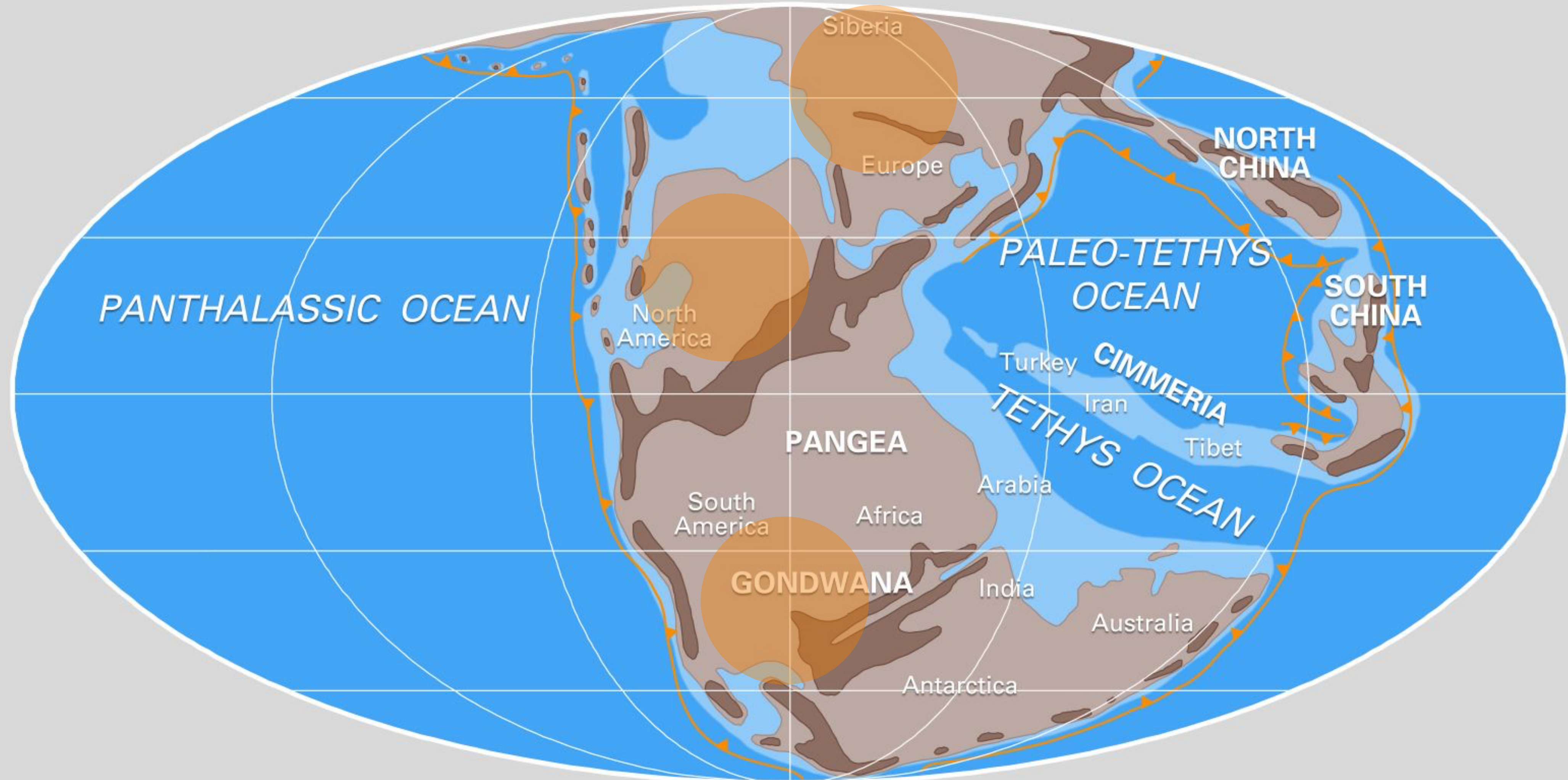


Mountains Land Shallow seas Deep ocean basins Subduction zone (triangles point in the direction of subduction)

SOURCE: © 2001 C.R. Scotese, PALEOMAP Project

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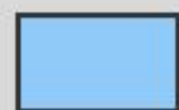
Early Triassic 237 million years ago



Mountains



Land



Shallow seas



Deep ocean basins

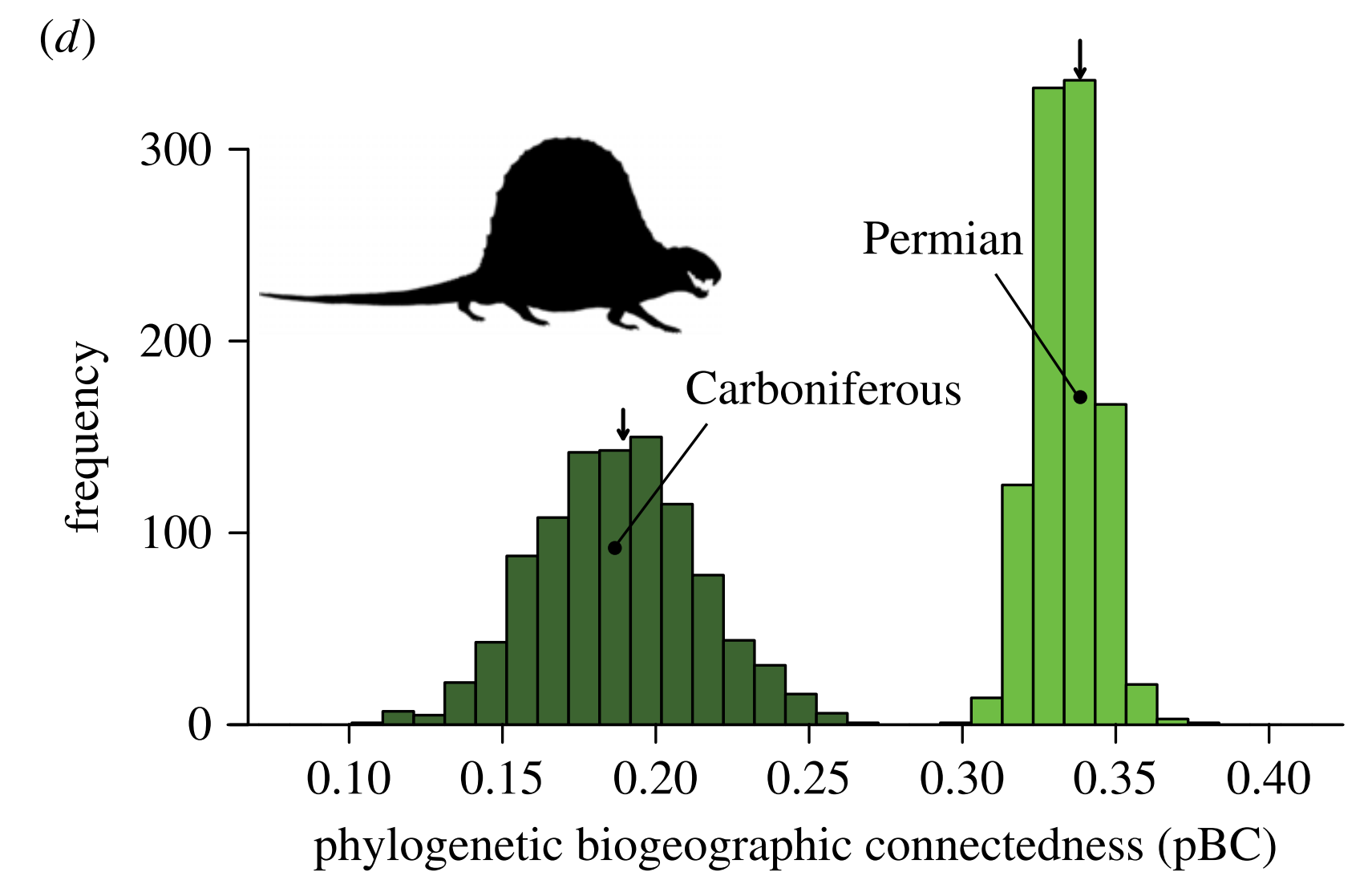
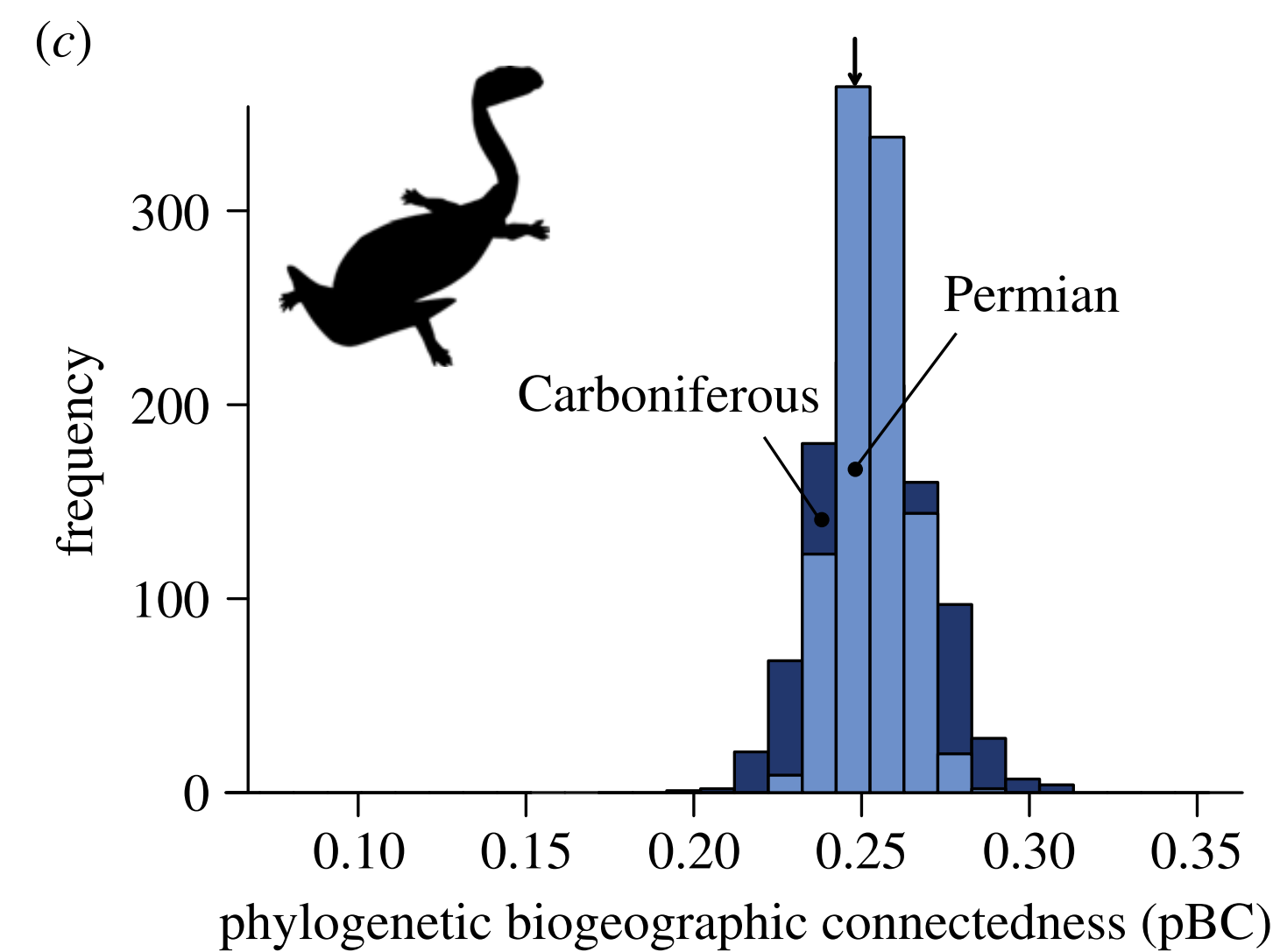
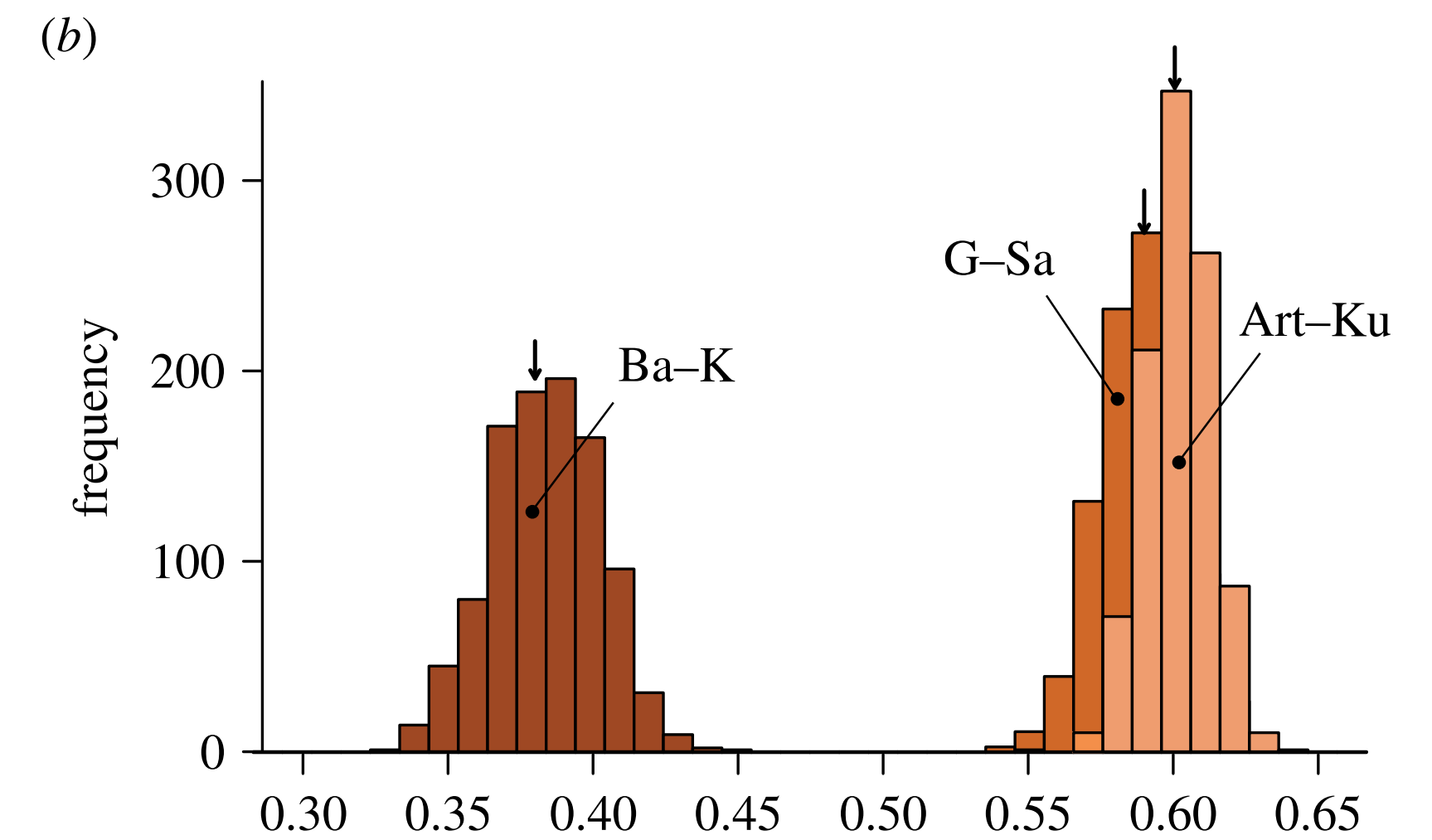
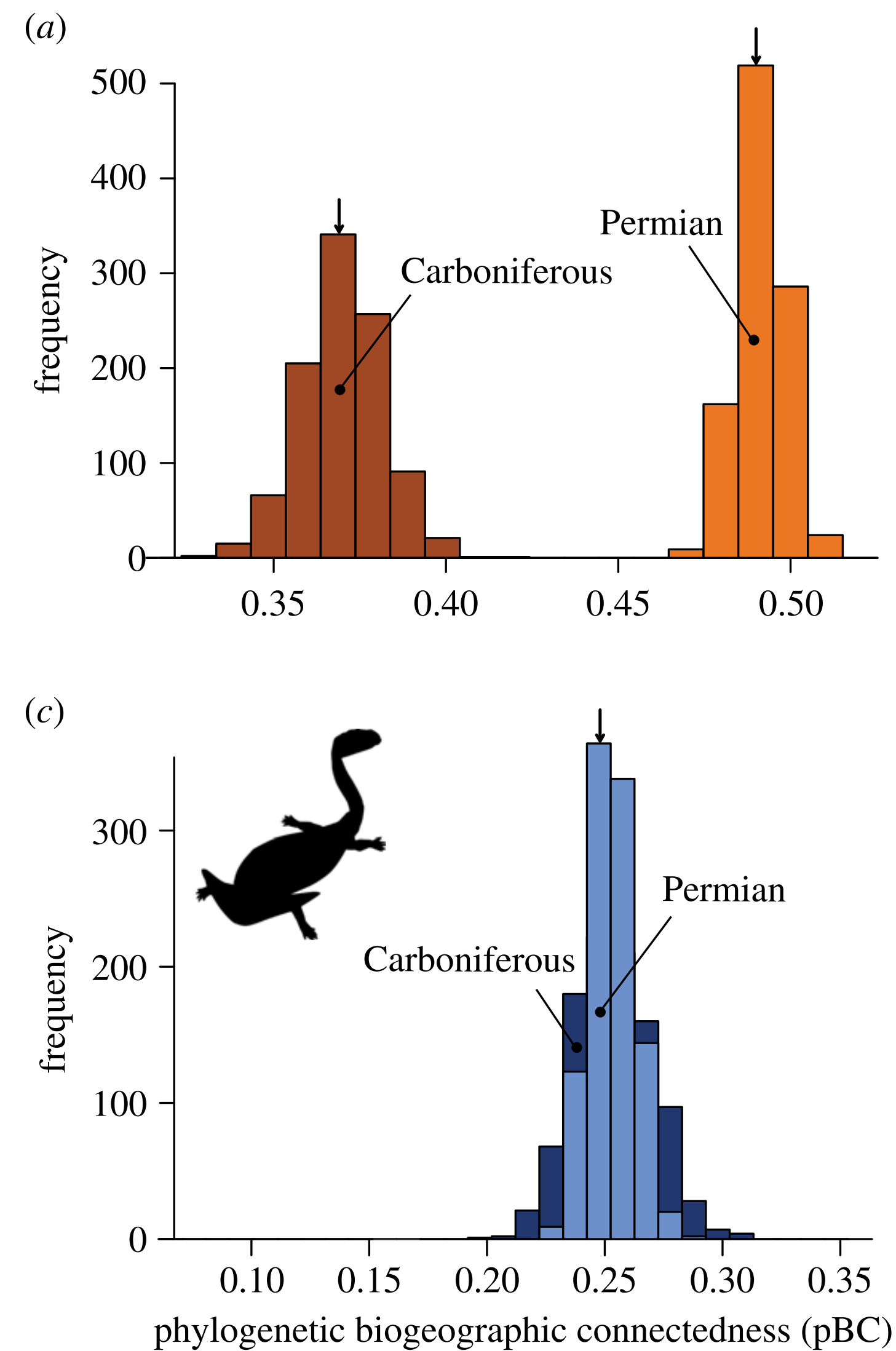


Subduction zone
(triangles point in the
direction of subduction)

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Increased
connectivity / migration
is supported by
different data and
methods



Dunne et al. ([2018](#))

BDMM-Prime checklist for macroevolution

We recommend always double checking your model parameters

Parameter	Notes
Birth (speciation) Rate — λ	usually estimated (time or lineage variable)
Death (speciation) Rate — μ	usually estimated (time or lineage variable)
Sampling (fossilisation) Rate — ψ	usually estimated (time or lineage variable)
Rho (extant sampling) Probability — ρ	usually fixed
Removal Probability — r	not applicable in macroevolution, fix to zero
Migration Rate — m	usually estimated (time or lineage variable)
Birth Rate Among Demes	not applicable in macroevolution, fix to zero